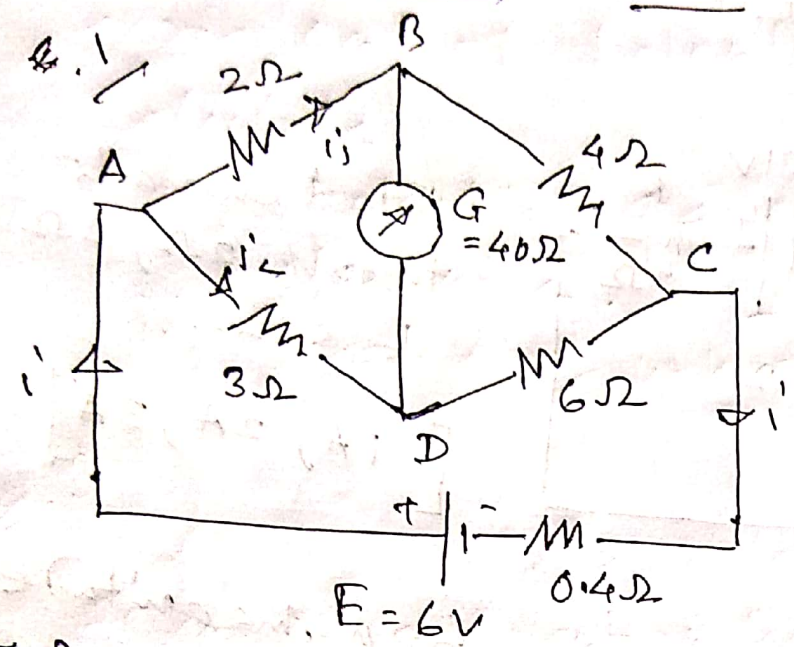
on Wheatstone & Metre Bridge

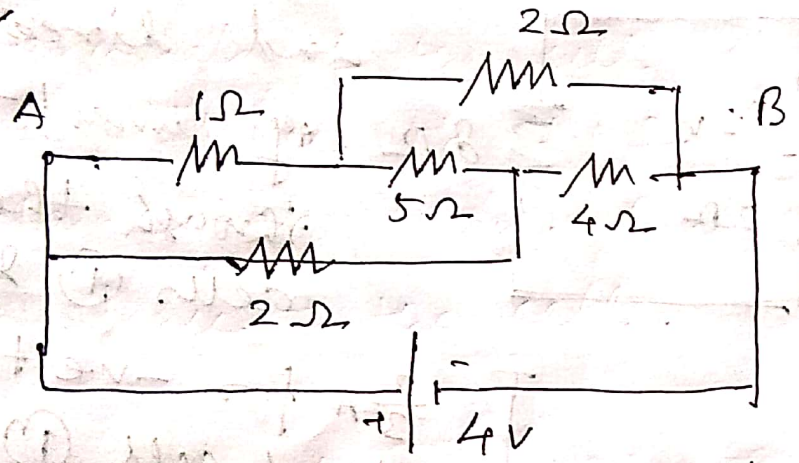
Q.1/



Find i_1, i_2 & i

[0.9A, 0.6A, 1.5A]

Q.2/



Calculate current drawn from the battery in this circuit?

[Ans 2A]

Q.3/ Three resistance P, Q, R each of 2Ω and an unknown resistance 'S' form the four arm of a wheatstone Bridge circuit. When a resistance 6Ω is connected in parallel to 'S' the bridge gets balanced. What is value of 'S'=?

[3Ω]

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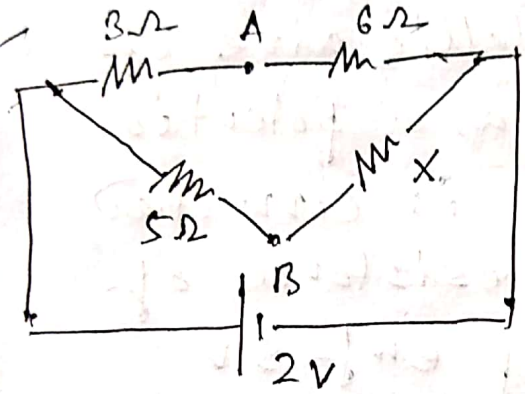
4. 

Find value of X for which A & B are at same potential?
[Ans 10Ω]

5. The left hand and right hand gaps of a metre bridge contain resistances of 4Ω and 2Ω resp. When 4Ω is shunted by a length of a wire, the balance point is found to be at 60 cm from left end of the bridge wire. What is resistance of the shunt? [12Ω]

6. Two unknown resistances are placed in the gaps of a metre bridge and the null point is found to be 30 cm from one end of the wire. When one of the resistances is shunted by 21Ω, the null point is 20 cm from the same end. Find value of the unknown resistances. [15Ω, 35Ω]

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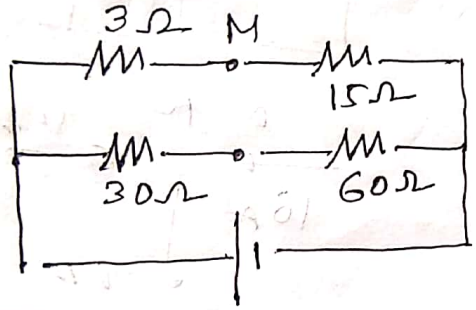
MB

7. A resistance of 2Ω is connected across one gap of a metre bridge (length = 100cm) & an unknown resistance (72Ω) is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20cm. Find the unknown resistance

[3Ω]

8.



How much resistance should be connected to 15Ω resistor (in f's), so that the points M & N are at same potential? (10Ω parallel to 15Ω)

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